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REVIEW

Sleep disorders in hepatocellular carcinoma

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Abstract

As one of the most prevalent malignant tumors, hepatocellular carcinoma (HCC) represents a major global public health burden. Traditionally, HCC pathogenesis has been attributed to chronic liver diseases (viral hepatitis, cirrhosis) and aflatoxin exposure. However, with evolving lifestyles and environmental changes, sleep disorders have become increasingly prevalent. Emerging evidence suggest that sleep disorders may contribute to hepatocarcinogenesis through multiple mechanisms, including immunity environment disorder, oxidative stress, metabolic dysregulation, disruption of gut microbiota, and circadian rhythm disruption, thereby influencing disease progression and patient prognosis. This review summarizes epidemiological evidence on the relationship between sleep disorders and HCC incidence, explores the underlying mechanisms through which sleep disorders contribute to HCC, and discusses clinical challenges and potential intervention strategies. Our objective is to provide novel insights into HCC prevention and therapeutic approaches.

Key Words: Sleep disorders; Circadian rhythm; Clock gene; Hepatocellular carcinoma; Risk factors; Prevention strategies

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Core Tip: Hepatocellular carcinoma (HCC) remains a globally prevalent malignancy, traditionally attributed to etiological factors such as viral hepatitis, alcohol-related liver disease, and metabolic dysfunction-associated fatty liver disease. Alarmingly, the global incidence of HCC continues to rise, with more than 900000 new cases reported in 2020, a trend not fully explained by these established risk factors. Emerging evidence highlights sleep disorders as a novel contributor to hepatocarcinogenesis. This review systematically explores the association between sleep disturbances and HCC pathogenesis, focusing on the underlying pathological mechanisms through which disrupted sleep patterns may drive HCC progression.

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INTRODUCTION

Hepatocellular carcinoma (HCC) is one of the most prevalent malignant tumors globally and represents a major health burden worldwide. According to the 2020 global cancer statistics, HCC accounts for over 900000 new cases and approximately 830000 deaths annually. Both its incidence and mortality rates have shown a significant upward trend over the past decade, particularly in East Asia and Africa[1].

Common etiological factors for HCC include viral hepatitis, alcohol-associated liver disease, and metabolic dysfunction-associated fatty liver disease (MAFLD). The global prevalence of MAFLD, closely linked to factors such as obesity and insulin resistance, continues to rise alongside changing lifestyles[2]. Concurrently, sleep disorders, as another unhealthy lifestyle factor, have been implicated in liver cancer development. Yang *et al*[3] investigated the relationship between various sleep traits and primary liver cancer (PLC), finding that adequate and regular sleep plays a significant protective role. Specifically, longer nocturnal sleep duration was significantly inversely associated with reduced PLC risk (β = -0.002, P = 0.026). Conversely, daytime napping (β = 0.001, P = 0.043) and insomnia (inverse-variance weighted P = 0.022) both demonstrated significant positive associations with increased HCC risk. Wu *et al*[4] found that obstructive sleep apnea (OSA) significantly increases liver cancer incidence (RR: 1.19, 95%CI: 1.10-1.29).

Furthermore, animal studies indicate that sleep disorders can exacerbate liver fibrosis progression, induce preneoplastic lesions, and indirectly elevate HCC risk through multiple pathways, such as promoting hepatic steatosis and insulin resistance, and aggravating hepatocellular DNA damage[5-7]. These findings suggest that sleep disorders may constitute a novel risk factor, warranting systematic investigation into their pathophysiological interactions with HCC development.

Currently, chronic insomnia symptoms affect approximately 25% of adults[8], while OSA impacts nearly 1 billion individuals globally[9]. Without intervention, these sleep disorder patients face a substantially elevated risk of developing liver cancer, posing a serious public health challenge. This review primarily examines the risk of HCC development in individuals with sleep disorders and the associated pathobiological mechanisms. It also explores the potential value of targeted sleep interventions for HCC primary prevention, along with current clinical challenges and future research directions.

ASSOCIATION BETWEEN COMMON SLEEP DISORDERS AND HCC DEVELOPMENT

According to the International Classification of Sleep Disorders, Third Edition, sleep disorders are categorized into seven major types[10]. Among these, insomnia, OSA, and circadian rhythm disruption are currently considered the most common sleep disorders associated with liver cancer. However, the epidemiological associations between other sleep disorders and the pathogenesis of HCC require further investigation, and scientifically robust evidence is needed to substantiate the potential relationships between different sleep disorders and liver cancer occurrence.

Insomnia is one of the most common sleep disorders contributing to liver cancer development. It is characterized by difficulty initiating sleep, frequent nocturnal awakenings, early morning awakening with an inability to return to sleep; and overall manifests as reduced sleep duration and poor sleep quality. Clinically, insomnia affects up to 80% of patients with chronic liver disease (*e.g.*, cirrhosis)[11], and the severity of their disease shows a significant correlation with worsening sleep disorders[12]. A prospective cohort study involving 356894 United Kingdom Biobank participants demonstrated that unhealthy sleep patterns, including insomnia, significantly increased the risk of HCC [hazard ratio (HR): 1.46, 95%CI: 1.15-1.85][13]. The NIH-AARP cohort study further revealed a U-shaped association between sleep duration and HCC risk, with both short (< 5 hours; HR: 2.00, 95%CI: 1.22-3.26) and long (≥ 9 hours; HR: 1.63, 95%CI: 1.04-2.65) sleep durations increasing risk[14]. Globally, insomnia symptoms affect 25%-30% of adults, with approximately 50% progressing to chronic insomnia[15]. Among Asian populations, the prevalence of insomnia in Chinese adults has

reached 15%, with higher rates observed in younger individuals[16], potentially attributable to lifestyle factors. Therefore, improving insomnia is crucial for preventing liver cancer development.

OSA is characterized by recurrent episodes of apnea and hypopnea during sleep, with a higher prevalence in middle-aged men (approximately 4%) compared to women (2%)[17]. Evidence suggests that OSA is not only a risk factor for liver cancer but may also exacerbate disease progression. A meta-analysis of 18 cohort studies indicated that OSA significantly increased liver cancer incidence (RR: 1.19, 95%CI: 1.10-1.29)[4]. Lin *et al*[18] found significant correlations between sleep-related hypoxemia in MAFLD patients and markers of hepatocellular injury (*e.g.*, elevated alanine aminotransferase) as well as hepatic steatosis indices. This confirms that the hypoxic state induced by OSA can directly cause liver injury and promote chronic liver disease progression. Qu *et al*[19] reported consistent findings, further demonstrating that hepatic hypoxia activates inflammatory pathways, leading to enhanced liver inflammation and oxidative stress, changes that may promote hepatocarcinogenesis.

Under normal conditions, the circadian rhythm system maintains homeostasis in numerous biological functions, including cell proliferation, metabolic balance, and immune regulation[20]. Desynchronization between the endogenous circadian clock and environmental light-dark cycles, caused by factors like shift work or transmeridian travel, disrupts organismal homeostasis (including sleep-wake patterns, metabolic processes, and hormone secretion), thereby increasing carcinogenic risk[21,22]. In 2007, shift work involving circadian disruption was classified as "probably carcinogenic to humans" (Group 2A)[23]. Clinical research by Wegrzyn *et al*[24] supported this, showing that long-term rotating night-shift work was associated with a higher risk of breast cancer (HR = 1.40, 95%CI: 1.00-1.97). Multiple clinical studies also indicate an elevated cancer risk among male and female flight attendants frequently crossing time zones, *e.g.*, skin cancer: Melanoma [standardised incidence ratio (SIR): 2.3, 95%CI: 1.7-3.0], non-melanoma (SIR: 2.1, 95%CI: 1.7-2.8), basal cell carcinoma (SIR: 2.5, 95%CI: 1.9-3.2)[25]. VoPham *et al*[26] also identified circadian disruption as a potential independent risk factor for liver cancer, reporting that within the same time zone, each 5-degree increase in longitude from east to west was significantly associated with increased HCC risk (incidence rate ratio: 1.07; 95%CI: 1.01-1.14, $P = 0.03$).

Collectively, these findings establish sleep disorders as independent risk factors for the development and progression of HCC as shown in Table 1. In clinical practice, enhanced sleep management for high-risk populations (*e.g.*, patients with chronic liver disease) is warranted to mitigate disease risk. Future research should further explore the impact of sleep interventions on liver cancer prognosis and elucidate the interactions between sleep disorders and other risk factors.

PATHOGENIC MECHANISMS UNDERLYING SLEEP DISORDER-ASSOCIATED HEPATOCARCINOGENESIS

With increasing health awareness in modern society and a rising prevalence of sleep disorders, research related to sleep disorders has garnered significant attention. Substantial evidence now demonstrates that sleep disorders can promote liver cancer development through multiple mechanisms. We will delineate these mechanisms, focusing on immune dysregulation, oxidative stress, metabolic disorders, gut dysbiosis, and circadian rhythm disruption (Figure 1).

Immunity environment disorder

Sleep disorders disrupt immune homeostasis. On one hand, they can induce persistent activation of pro-inflammatory cytokines, fostering a chronic inflammatory microenvironment within the liver. On the other hand, sleep disorders impair immune cell activity, diminishing their capacity to recognize and eliminate nascent cancer cells. Collectively, these effects contribute to the development of HCC. Firstly, sleep disorders amplify inflammatory-immune crosstalk, accelerating HCC progression[14]. Chronic infection with hepatitis B virus (HBV) or hepatitis C virus inherently causes sustained hepatocyte damage and inflammatory responses, rendering hepatocytes more susceptible to malignant transformation [27-29]. Concomitant sleep deprivation further elevates systemic levels of pro-inflammatory cytokines [*e.g.*, tumor necrosis factor-α (TNF-α), interleukin (IL)-6][30]. Existing research has also identified that inflammatory mediators within the tumor microenvironment (TME), such as IL-6 and β-arrestin 1, persistently activate the NF-κB and STAT3 signaling cascades. This activation directly promotes tumor cell proliferation while simultaneously inhibiting apoptosis[31,32]. Preclinical evidence indicates that sleep deprivation upregulates hepatic lipase activity, thereby promoting hepatic steatosis and insulin resistance[5]. These complications synergistically exacerbate hepatic lipid accumulation, perpetuating the inflammatory cascade[33]. Secondly, sleep disorders impair tumor immune surveillance by disrupting immune cell effector functions and cytokine networks[34]. Huang *et al*[35] demonstrated that sleep deprivation in mice leads to reduced numbers and impaired function of T lymphocytes and natural killer cells in peripheral circulation. Concurrently, it increases the infiltration of immunosuppressive CD11b+ cells from the spleen into the TME *via* peripheral blood. Notably, sleep-related immune dysregulation extends beyond peripheral circulation to lymphoid organs. Zager *et al*[36] found that sleep disruption causes a generalized reduction across all splenocyte subsets (primarily T and B cells). Furthermore, sleep disorders disrupt the Th1/Th2 homeostasis, which is critical for immune balance. This results in Th2 cell hyperfunction and the secretion of cytokines such as IL-4 and IL-10. These cytokines promote tumor angiogenesis while simultaneously suppressing Th1 cell function. This dual action weakens the host's anti-tumor immune response, enabling tumor cells to evade immune surveillance, proliferate unchecked, and establish an immune escape microenvironment, thereby further promoting HCC development[37].

Oxidative stress

Oxidative stress refers to a pathological state characterized by an imbalance between reactive oxygen species (ROS)

Table 1 Impacts of different types of sleep disorders on liver cancer

Ref.	Country	Type of sleep disorders	Type of cancer	Outcomes
[13]	United Kingdom	Insomnia	HCC	HR: 1.46, 95%CI: 1.15-1.85
[14]	United States	Insomnia	HCC	HR: 2.00, 95%CI: 1.22-3.26
		Long sleep durations	HCC	HR: 1.63, 95%CI: 1.04-2.65
		Daytime napping	HCC	HR: 1.46, 95%CI: 1.04-2.06
[4]	China	Obstructive sleep apnea	Liver cancer	RR: 1.19, 95%CI: 1.10-1.29
[26]	United States	Circadian rhythm disruption	HCC	IRR: 1.07; 95%CI: 1.01-1.14

Summary of epidemiological studies investigating associations between specific sleep disorders and liver cancer risk, categorized by disorder type, geographic region, and reported effect measures (hazard ratios, relative risks, or incidence rate ratios with 95%CI): Insomnia, prolonged sleep duration, daytime napping, obstructive sleep apnea and circadian rhythm disruption increase the risk of liver cancer. HR: Hazard ratio; RR: Relative risk; IRR: Incidence rate ratio; HCC: Hepatocellular carcinoma.

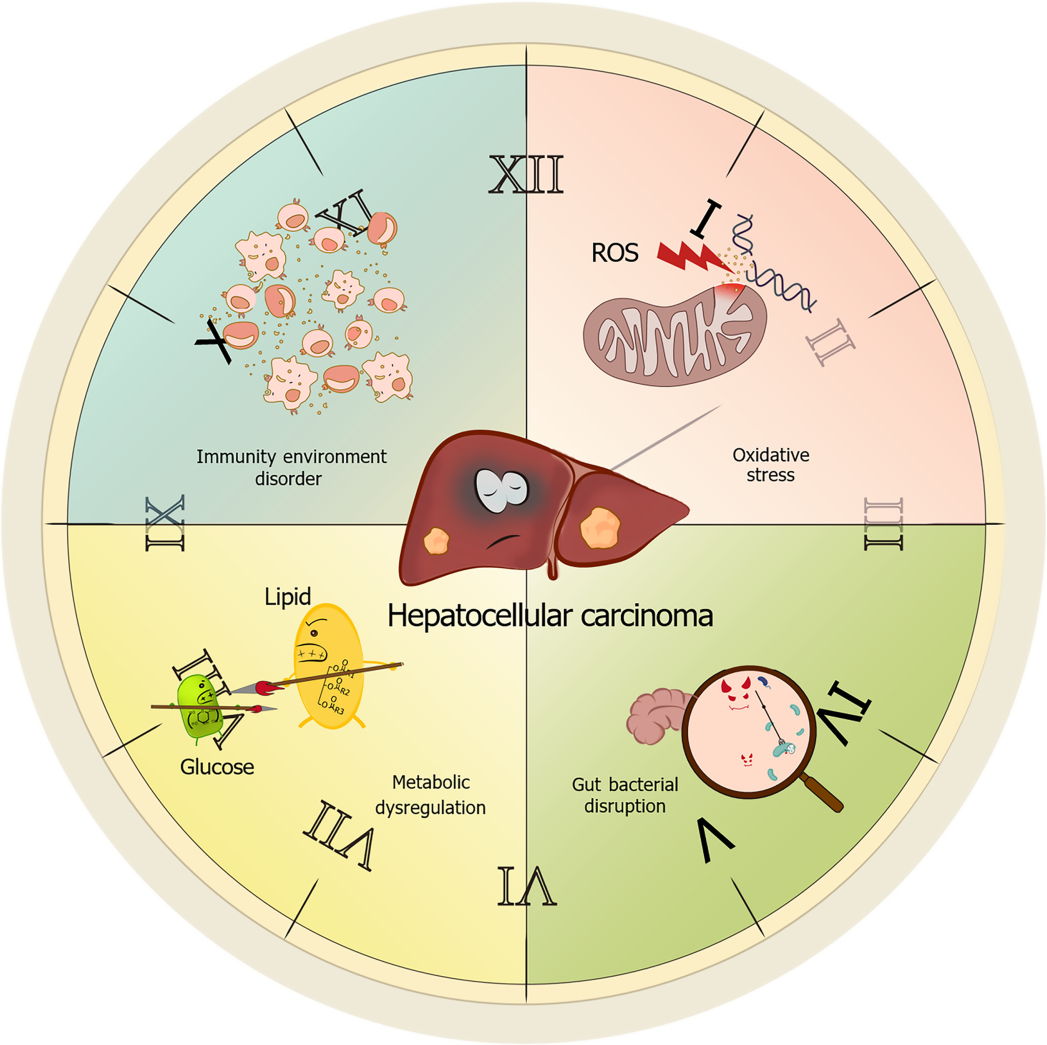

Figure 1 Pathogenic mechanisms underlying sleep disorder-associated hepatocarcinogenesis. This diagram illustrates the key mechanisms that lead to the occurrence of liver cancer: The immune environment disorder reflects the strengthening of chronic inflammatory response and the weakening of anti-tumor monitoring. The destruction of intestinal bacteria can cause ecological disorders, promote metabolic and immune dysfunction. Metabolic disorders manifest as changes in lipid and glucose homeostasis. Oxidative stress amplifies cellular damage through reactive oxygen species. These interrelated pathways synergistically drive the progression of hepatocellular carcinoma. ROS: Reactive oxygen species.

production and antioxidant defense systems under endogenous or exogenous stimuli, leading to oxidative damage accumulation. Mitochondrial dysfunction constitutes the predominant mechanism underlying this process[38-40]. Excessive mitochondrial-derived ROS disrupts lipid bilayer integrity, induces DNA strand breaks, and promotes protein carbonyl modifications, ultimately triggering apoptosis or malignant transformation[41]. Oxidative stress serves as a critical pathogenic nexus in HCC, where sleep disorders exacerbate hepatic redox imbalance through mitochondrial dysfunction and impaired antioxidant defense, thereby accelerating hepatocarcinogenesis *via* ROS-mediated genomic instability and pro-tumorigenic signaling.

Preclinical models further confirm that sleep deprivation leads to the accumulation of ROS in the intestine, which can ultimately be fatal without antioxidant intervention[42,43]. Within the hepatic context, sustained ROS assault on hepatocytes triggers chronic inflammation, activates hepatic stellate cells (HSCs), and accelerates fibrosis[44,45]. Furthermore, oxidative stress induces cell cycle arrest by interfering with the function of checkpoint kinase 2[46-48] and increases genomic instability, thereby promoting tumorigenesis[49].

Sleep disorders profoundly disrupt redox homeostasis. Multiple studies indicate that chronic sleep disorders impair mitochondrial dynamics, leading to compromised mitochondrial function and a further exacerbation of ROS generation [50,51]. Previous research has established that the Kelch-like ECH-associated protein 1-nuclear factor erythroid 2-related factor 2 (Nrf2) pathway is a crucial defense mechanism protecting the liver against oxidative stress damage[52]. Concurrently, Nrf2 serves as a key transcription factor in maintaining cellular redox homeostasis during the sleep-wake cycle [53]. Mechanistically, sleep disorders may suppress Nrf2 nuclear translocation, downregulate antioxidant gene expression, and impair ROS scavenging capacity, collectively promoting hepatocarcinogenesis. Relevant clinical studies also corroborate the link between oxidative stress and liver cancer. Oxidative stress-induced autophagosome formation promotes extensive HBV replication, thereby increasing the risk of HCC recurrence[54]. Additionally, elevated ROS levels in patients with chronic hepatitis B suppress the expression of suppressor of cytokine signaling 3 *via* Snail-mediated epigenetic silencing. This suppression results in sustained activation of the IL-6/STAT3 signaling pathway, ultimately leading to HCC development[55]. In summary, oxidative stress drives hepatocarcinogenesis through ROS-mediated DNA damage, inflammation activation, and dysfunction of protein inhibitory mechanisms[56,57]. As a key instigator, sleep disorders amplify hepatic oxidative damage through mitochondrial dysfunction, suppression of antioxidant defenses, and enhancement of pro-carcinogenic signaling, thereby establishing a pathogenic "sleep disorder-oxidative stress-HCC" axis.

Metabolic dysregulation

Hepatic metabolic dysfunction - centered on insulin resistance - constitutes a critical pathological foundation for HCC, with sleep disorders exacerbating hepatocarcinogenesis through disruption of glucose/Lipid homeostasis, aberrant energy sensing, and circadian misalignment.

The liver, acting as the central hub for glucose and lipid metabolism, is susceptible to metabolic dysfunction, which underpins the pathogenesis of HCC. Insulin resistance drives this process by disrupting glucose and lipid homeostasis: Under physiological conditions, insulin stimulates hepatic glucose uptake and glycogen synthesis while inhibiting gluconeogenic gene expression *via* the PI3K/AKT pathway[58,59]. During insulin resistance, reduced hepatocyte insulin sensitivity impairs GLUT4 membrane translocation, diminishes glycogen synthesis, and compromises AKT/FOXO1-mediated suppression of gluconeogenesis, thereby increasing hepatic glucose output. This persistent hyperglycemia provides an energy substrate for tumorigenesis[60-64]. Clinically, type 2 diabetes mellitus promotes HCC development through IGF-1-mediated inhibition of apoptosis, fostering hepatocyte hyperproliferation[65]. Concurrently, insulin resistance increases pathological free fatty acid uptake and triglyceride synthesis, inducing hepatic steatosis. The generation of excess ROS, diacylglycerol, and ceramides further impairs insulin signaling through a positive feedback loop. The synergy between lipid accumulation and mitochondrial oxidative stress establishes a vicious cycle that accelerates the progression of metabolic liver disease and significantly elevates HCC risk[66,67].

Sleep disorders profoundly amplify these metabolic derangements: Animal studies demonstrate that sleep-deprived mice develop hepatic steatosis and insulin resistance, exhibiting up to a 67.9% increase in hepatic triglycerides and upregulation of lipogenic genes (*e.g.*, Elovl3, Lipin1)[5,68]. Metabolic homeostasis in young adult mice is more vulnerable to sleep deprivation than in older mice, subsequently inducing aging-like features[69]. Furthermore, phosphatidylinositol-5-phosphate 4-kinase type 2 gamma-deficient cells exhibit heightened sensitivity to insulin-mediated PI3K/AKT signaling and exploit the insulin-rich hepatic environment for organ-specific metastasis, thereby increasing the burden of cancer liver metastases[70]. In an experimental model investigating spontaneous liver cancer formation, circadian disruption was found to activate the constitutive androstane receptor, accelerating MAFLD-associated HCC development by promoting increased lipid synthesis, disruption of peripheral clocks, and sympathetic nervous system dysfunction[71].

In summary, sleep disorders disrupt glucose and lipid metabolic homeostasis, activate oncogenic pathways, and ultimately drive carcinogenesis through tripartite mechanisms: Insulin resistance, aberrant energy sensing, and circadian disruption. Future research should further explore the long-term impact of sleep disorders on metabolic pathways to optimize clinical practice.

Disruption of gut microbiota

Sleep disorders exacerbate microbial dysregulation through circadian- neuroendocrine crosstalk, while gut-derived inflammatory/metabolic signals reciprocally drive HCC progression. The core circadian regulator, brain and muscle ARNT-like 1 (BMAL1), governs diurnal oscillations of gut microbial composition. Circadian disruption downregulates intestinal BMAL1 expression, reducing beneficial genera (*Bacteroides*, *Lactobacillus*) while enriching pathobionts (*Helicobacter*, *Enterobacteriaceae*), thereby instigating dysbiosis-driven inflammation[72,73]. Preclinical models reveal that sleep deprivation elevates serum corticotropin-releasing hormone, adrenocorticotropic hormone, and corticosterone

levels, indicating hypothalamic-pituitary-adrenal (HPA) axis hyperactivation. Concurrently, sleep-deprived rodents exhibit an increased abundance of opportunistic pathogens, suggesting that sleep disorders may contribute to gut microbial dysbiosis *via* HPA axis activation. When the gut microbiota is imbalanced, it can lead to the breakdown of the intestinal mucosal barrier, causing lipopolysaccharide (LPS) to enter the portal vein system. Elevated hepatic LPS concentrations trigger Toll-like receptor 4 (TLR4) signaling in Kupffer cells and HSCs. This activation stimulates excessive release of inflammatory factors, driving hepatic inflammation and oxidative injury while suppressing apoptosis and inducing DNA lesions[74]. Simultaneously, LPS engages TLR4 receptors on liver endothelial cells, initiating the MyD88-dependent TLR cascade. Through extracellular protease modulation, MyD88 governs angiogenesis - a pivotal process in HCC development - and facilitates cirrhosis-to-HCC transition[75]. Furthermore, LPS provokes hepatic progenitor cells (HPCs) to differentiate into myofibroblasts. Secreted IL-6 and TNF-α from these cells activate oncogenic Ras pathways and inactivate p53 tumor suppressor signaling in HPCs, ultimately accelerating aberrant HPC proliferation, malignant transformation, and HCC initiation[76,77]. Gut microbiota dysbiosis disrupts hepatic bile acid metabolism, elevating toxicity. Such imbalance promotes liver tumor advancement *via* the BA-CXCL16-CXCR6 axis[78]. In addition, gut microbiota can indirectly lead to disease by stimulating inflammatory states, such as cancer inflammation induced by enterotoxigenic fragile bacteria developing through STAT3- and Th17-dependent pathways[79,80], while inflammatory states also increase the production of ROS, inducing DNA damage and promoting tumor progression[81-83].

PREVENTIVE AND THERAPEUTIC STRATEGIES TARGETING SLEEP DISORDER-ASSOCIATED HEPATOCARCINOGENESIS

Currently, targeted sleep interventions for high-risk HCC patients face several clinical challenges. Firstly, patients with chronic liver disease often present with multiple comorbidities and complex health conditions. Certain sedative-hypnotic drugs may adversely affect hepatic function[84], imposing an additional burden on an already compromised liver. Consequently, careful consideration of both drug safety and efficacy is essential[85]. Secondly, HCC treatments, including surgery, chemotherapy, and radiotherapy, may induce or exacerbate sleep disorders[86-88]. Concurrently, treatment-related side effects such as pain and fatigue can further disrupt sleep[89], complicating the management of sleep disorders. Additionally, medications used to manage these symptoms may potentially interact with sleep-targeting agents. Furthermore, psychological factors cannot be overlooked in sleep management. HCC patients frequently experience comorbid anxiety and depression, which can significantly worsen sleep disorders[90]. Implementing effective psychological interventions requires specialized mental health professionals; however, resource constraints, including personnel shortages in clinical practice, often hinder optimal management.

Therefore, individualized treatment represents a key future direction. Personalized management plans must be developed through comprehensive consideration of patient-specific factors, including genetic background, disease progression, sleep disorder subtype, medication adherence, and tolerance levels.

Healthy sleep behaviors

Maintaining consistent sleep-wake schedules constitutes a critical component of circadian rhythm management. Establishing fixed bedtimes and wake-up times facilitates circadian entrainment, thereby preserving normal physiological rhythms[91,92]. Chronic sleep deprivation disrupts circadian homeostasis, precipitating endocrine dysfunction, metabolic dysregulation, and elevated risks of sleep disorders and HCC[71]. For individuals engaged in transmeridian travel or shift work, proactive circadian adaptation strategies are essential to mitigate jet lag-induced chronodisruption[93]. Empirical evidence indicates that adherence to 7-8 hours of nocturnal sleep duration establish robust physiological defenses through maintenance of circadian integrity and optimal physiological functioning[14].

Ambient factors including photic exposure, thermal conditions, and acoustic stimuli significantly influence sleep architecture. Blue light, a high-energy visible spectrum emitted by electronic devices (*e.g.*, smartphones, computers, and tablets), exerts melatonin-suppressive effects *via* retinal photoreceptor activation. Given melatonin's pivotal role in sleep initiation, such photic suppression correlates with prolonged sleep latency and diminished sleep efficiency[94]. Implementing digital device abstinence 1-2 hours pre-sleep or utilizing blue light filtering eyewear represents effective countermeasures[95,96]. Thermal optimization within 18-22 °C enhances sleep continuity through autonomic thermoregulation, whereas thermal deviations promote sleep fragmentation[97]. Noise is also an important factor affecting sleep. The noise level in the sleeping environment should be controlled below 30 decibels[98]. Earplugs, soundproof curtains, and other devices can be used to reduce noise interference and create a quiet sleeping environment.

Pharmacotherapy

Pharmacotherapy demonstrates clinical efficacy in managing sleep disorders (Table 2). The current pharmacological armamentarium primarily comprises non-benzodiazepine receptor agonists (Z-drugs), benzodiazepines (BZDs), melatonin and melatonin receptor agonists, sedating antidepressants, and orexin receptor antagonists (ORAs). BZDs (estazolam) potentiate γ-aminobutyric acid (GABA) ergic neurotransmission through GABA receptor modulation, effectively ameliorating prolonged sleep latency and sleep maintenance difficulties[99]. However, their clinical utility is constrained by dependence liability, tolerance development, and residual sedation, necessitating rigorous risk-benefit analysis[100,101]. Z-drugs (*e.g.*, zopiclone, zolpidem) share similar GABAergic mechanisms but demonstrate rapid onset, abbreviated elimination half-lives, and reduced adverse effect profiles, rendering them preferable for short-term insomnia management[102,103]. In hepatopathy patients with anxiety or depression, antidepressants (*e.g.*, mirtazapine, trazodone)

Table 2 Comparison of common sleep disorders medications

Ref.	Category	Advantages	Limitations	Target population
[102,103]	Non-benzodiazepines	Rapid onset, lower risk of dependence, minimal next-day residual effects	No anxiolytic efficacy; sleepwalking risk	Short-term insomnia, elderly patients
[99-101]	Benzodiazepines	Potent anxiolytic effects; well-established efficacy	High dependence risk; cognitive impairment; fall risk	Anxiety-related insomnia (short-term use)
[106,107]	Melatonin receptor agonists	No dependence risk; favorable safety	Limited efficacy in primary insomnia	Circadian rhythm disorders; comorbid respiratory conditions
[104,105]	Antidepressants	Effective for mood-related insomnia	Diverse side effects (*e.g.*, weight gain, hypotension)	Chronic insomnia with depression/anxiety
[108,109]	Orexin receptor antagonists	Novel mechanism; fewer adverse effects	Limited long-term efficacy data	Treatment-resistant insomnia; patients requiring prolonged therapy

Key characteristics of common insomnia medications are summarized, including: Non-benzodiazepines: Rapid onset with lower risk of dependence but limited anxiolytic effects. Benzodiazepines: Potent anxiolytic effects yet high dependence risk. Melatonin receptor agonists: No dependence risk yet limited efficacy in primary insomnia. Antidepressants: Preferred for comorbid mood disorders despite side effects. Orexin receptor antagonists: Novel mechanism and fewer adverse effects (long-term safety pending).

exert dual therapeutic effects by modulating monoaminergic transmission, concurrently improving sleep architecture and affective symptoms[104,105]. Melatonin and its receptor agonists constitute effective therapeutics for circadian rhythm sleep-wake disorders[106,107]. ORAs represent a novel therapeutic class for sleep disorders, exerting their therapeutic effects by modulating the physiological sleep-wake cycle. Clinical evidence demonstrates their capacity to significantly reduce sleep onset latency and wake after sleep onset duration, thereby promoting sleep patterns that more closely resemble natural physiological sleep. Notably, these agents exhibit superior efficacy compared to placebos and traditional medications such as zolpidem, while demonstrating significantly lower risks of addiction and withdrawal reactions relative to BZDs[108]. The favorable cognitive profile, characterized by minimal impact on daytime cognitive function, renders them particularly suitable for elderly populations and patients with intolerance or dependency risks associated with conventional therapies[109]. However, current clinical research remains insufficient in addressing the heterogeneity of therapeutic responses and establishing comprehensive long-term safety profiles. Clinical implementation requires careful consideration of individual patient characteristics, necessitating a balanced evaluation of therapeutic benefits *vs* potential risks to achieve precision medicine objectives in sleep disorders management.

Notably, pharmacotherapy in cirrhotic populations requires meticulous evaluation of hepatic metabolism pathways and potential drug-drug interactions to optimize the therapeutic index.

Non-pharmacological interventions

Cognitive behavioral therapy for insomnia (CBT-I), endorsed as a first-line therapeutic modality in European insomnia guidelines, demonstrates robust efficacy in managing chronic insomnia[99,110]. This multimodal intervention comprises three core components: Sleep hygiene optimization, stimulus control therapy (SCT), and sleep restriction therapy (SRT)[111]. Sleep hygiene education provides structured guidance on sleep-conducive behaviors, emphasizing circadian alignment through fixed sleep-wake schedules, avoiding the use of electronic devices before bedtime, and avoiding activities unrelated to sleep in the bedroom[112]. SCT employs operant conditioning principles to reinforce bed-sleep associations, mandating: (1) Bedtime initiation only upon sleepiness onset; (2) Conditioned arousal reduction through immediate bedroom exit following 15-20 minutes of wakefulness; and (3) Re-initiation of sleep attempts exclusively in drowsy states[113]. SRT systematically curtails time-in-bed to enhance homeostatic sleep pressure, thereby improving sleep efficiency metrics[114]. Research has shown that CBT-I can effectively improve the sleep quality of insomnia patients, reduce the occurrence of insomnia symptoms, and improve their quality of life[115,116].

For patients diagnosed with OSA, continuous positive airway pressure (CPAP) remains the gold-standard intervention. CPAP apparatus delivers titrated positive pressure *via* nasal/oral interfaces during sleep, maintaining upper airway patency through pneumonic splinting. Clinical studies demonstrate CPAP's multifaceted benefits: (1) Normalization of apnea-hypopnea index; (2) Attenuation of nocturnal hypoxemia; and (3) Deceleration of hepatic fibrogenesis progression; ultimately reducing HCC incidence through hypoxia inducible factor-mediated pathway modulation[117].

Emerging evidence indicates that structured exercise regimens demonstrate therapeutic efficacy in sleep disorder management. Moderate-intensity aerobic modalities (*e.g.*, ambulation, jogging, yoga) enhance cardiorespiratory fitness and promote autonomic nervous system equilibrium, thereby facilitating sleep homeostasis. A randomized controlled trial in oncological populations revealed that protocolized aerobic exercise significantly ameliorates sleep architecture parameters and improves health-related quality of life metrics[118]. Complementary and alternative medicine approaches, including acupuncture and therapeutic massage, exhibit clinical potential in sleep pathology management[119,120]. Mechanistic studies suggest massage therapy attenuates musculoskeletal hypertonicity and nociceptive signaling through somatosensory afferent modulation, consequently enhancing sleep continuity[121]. In HCC cohorts, such non-pharmacological interventions demonstrate dual benefits: Sleep optimization and psychological distress alleviation through cortisol rhythm normalization.

Deep brain stimulation (DBS), a stereotactic neurosurgical intervention utilizing high-frequency electrical modulation of target nuclei, has established therapeutic superiority in movement disorders including Parkinson's disease and essential tremor[122,123]. Recent advances in sleep-wake circuitry elucidation have propelled DBS applications into circadian dysregulation domains. DBS's unique advantages-stereotactic precision, reversible neuromodulation, and dose-titratable intervention-position it as a paradigm-shifting therapeutic strategy for refractory sleep disorders.

CONCLUSION

Overall, sleep disorders play an extremely important role in the occurrence and development of liver cancer, and there is a complex and closely related relationship between the two. Sleep disorders, such as insomnia, OSA-hypopnea syndrome, and circadian rhythm disorders significantly increase the risk of liver cancer through multiple mechanisms such as chronic inflammation, oxidative stress, metabolic dysregulation, immune suppression, disruption of gut microbiota, and clock gene dysregulation. All these lead to complex physiological and pathological changes, which include immune inflammation imbalance weakening the body's defense, DNA damage repair obstruction increasing the risk of gene mutations, biological clock gene loss leading to metabolic disorders, and liver fibrosis and cirrhosis accelerating deterioration of the liver microenvironment. While some progress has been made in evaluating the relationship between sleep disorders and liver cancer, there are still many issues that need to be addressed. Future research should be oriented to conduct large-scale, prospective cohort studies to unravel the causal factors, to identify the differential effects of different subtypes, and to explore the molecular mechanisms by which sleep disorders affect the occurrence and development of liver cancer. These would lead to identification of more potential targets, thereby providing a theoretical basis for developing new treatment strategies. Additionally, the development of precise interventions based on circadian rhythms, will become one of the key directions for future research. These studies are expected to open up new venues for early diagnosis and prevention, and individualized treatment, to reduce the incidence rate and mortality of liver cancer, improve the quality of life and survival rate of patients, and finally achieve the goal of translational medicine of "from sleep management to liver cancer prevention".

FOOTNOTES

Author contributions: Qian J and Ruan MH contributed equally to this work; Liu H conceived and designed the overall scope, structure, and central hypothesis of this review article; provided continuous scientific oversight and guidance throughout all stages of the project (conception, literature search strategy, drafting, and revision); secured and administered the financial support for the work; performed critical revisions of multiple drafts of the manuscript, focusing on the coherence of the narrative, the integration of key findings on sleep disorders (*e.g.*, sleep apnea, insomnia) and their mechanistic links to hepatocellular carcinoma, and the clinical implications; provided final approval of the version to be published; serves as a primary contact for communication with the journal. Feng XC conceived and designed the overall scope, structure, and central hypothesis of this review article; provided continuous scientific oversight and guidance throughout all stages of the project (conception, literature search strategy, drafting, and revision); performed critical revisions of multiple drafts of the manuscript, with a focus on the methodological rigor of the literature review, the accuracy of the pathophysiological mechanisms linking sleep disturbances to hepatocarcinogenesis (*e.g.*, circadian disruption, inflammation, metabolic dysregulation), and the strength of the epidemiological evidence; provided final approval of the version to be published; serves as a primary contact for communication with the journal and handles post-submission inquiries and revisions; Qian J and Ruan MH conducted the literature review and drafted the original manuscript; Jia JY and Ruan MH created the artwork; Feng XC Wang ZJ, Dong W, made critical revisions; all authors prepared the draft and approved the submitted version.

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